A Structural Selection Framework for Nonlinear Field Systems

Coherence, Entropy, and Instability in Dynamical Field Configurations

Structural Selection & CCI Series (2026)

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Overview

This research series develops a unified framework for understanding which structures persist in nonlinear field systems and why. The central idea is:

> *Not all mathematically valid configurations are dynamically realised. A structural selection principle determines which configurations persist.*

The framework combines:

- nonlinear field dynamics
- maximum entropy selection
- structural free energy
- information-theoretic diagnostics

At its core lies the **Critical Coherence Index (CCI)**:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\text{instability} \times \text{activity}}{\text{coherence}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}\,(1+\kappa\,U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}}+\Gamma_{\mathrm{cons}}+\Gamma_{\mathrm{corr}}+\Gamma_{\mathrm{int}}+\varepsilon}$$

Empirical scaling relation (Paper 07):

$$\boxed{\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \quad \alpha \approx 2.5 \text{ (1D)},\ \alpha \approx 3.0 \text{ (2D, within tested range)}}$$

$$r_s = 0.734 \text{ (1D)}, \quad r_s = 0.870 \text{ (2D)}$$

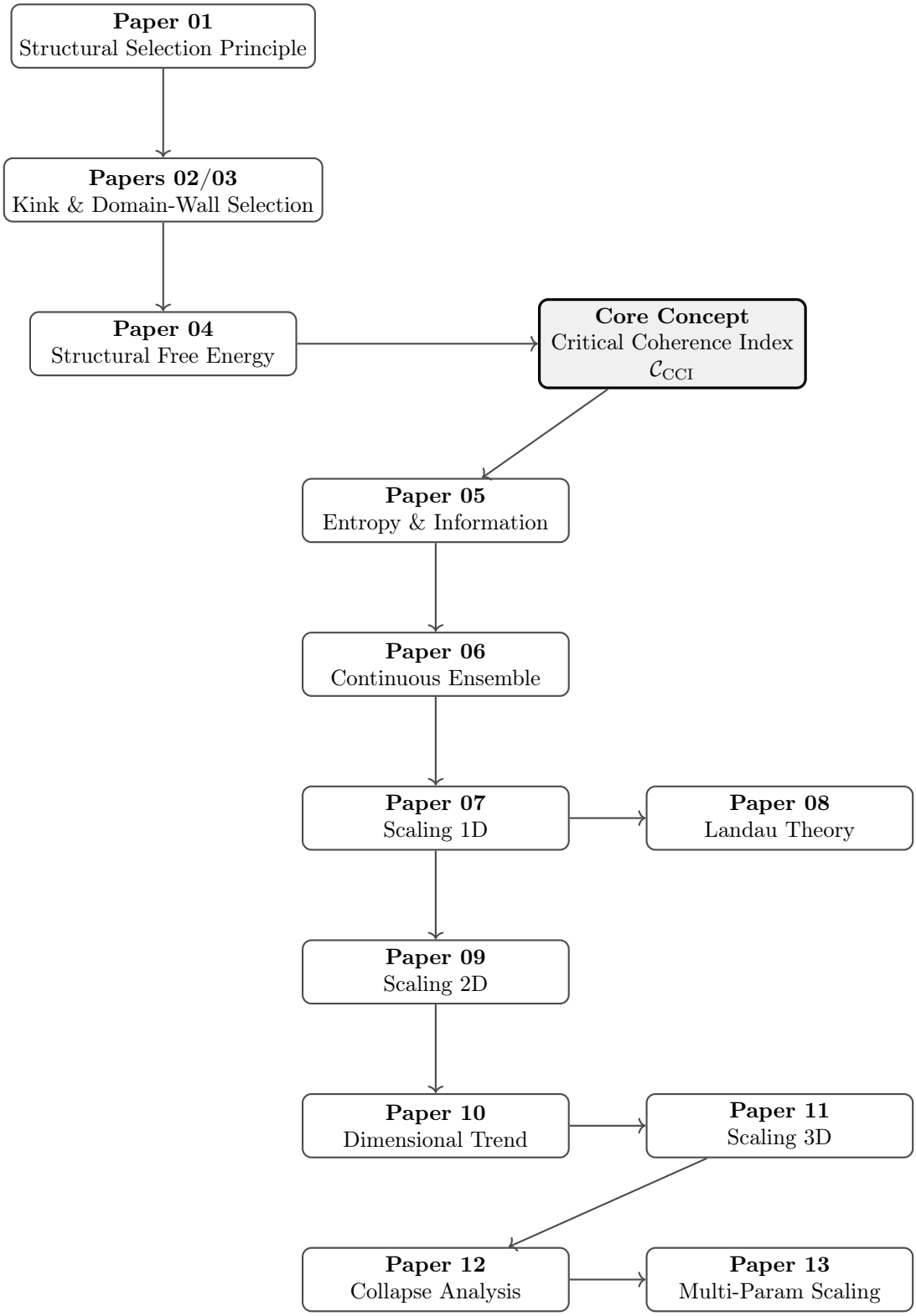
This relation is empirical and currently lacks a first-principles derivation. The exponents are obtained from ensemble statistics and are not claimed as fundamental constants.

The three-level interpretive hierarchy of the framework:

$$\underbrace{\mathcal{C}_{\mathrm{CCI}}}_{\text{instability index}} \quad \leftarrow \quad \underbrace{\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}}_{\text{empirical scaling proxy}} \quad \subset \quad \underbrace{\mathcal{F}_{\mathrm{struct}}}_{\text{effective selection functional}}$$

This hierarchy suggests that entropy–information scaling is not a fundamental law, but an emergent projection of a more general structural selection principle.

Paper Series Map



Emergent Picture:

Structure $\leftrightarrow$ Information $\leftrightarrow$ Stability

Paper Series

All papers are available at:

https://kriegchristof.academia.edu/research

1. **Paper 01 — Structural Selection Principle**
   Introduces the core hypothesis: maximum-entropy weighting over physical solution spaces.

$$P[\Phi] \propto \exp(-\beta\, \mathcal{F}_{\text{struct}}[\Phi]), \qquad \mathcal{M}_S = \{\Phi \mid \delta S[\Phi] = 0\}$$

2. **Papers 02/03 — Structural Selection in Nonlinear Field Theories**

   Kink selection in 1D (81/81 settings, weight 99.96%) and domain-wall selection in 2D (transition at $s_\star^c \in (0.20, 0.42)$).

3. **Paper 04 — Structural Free Energy**

   Defines coherence diagnostics, structural free energy, and regime classification via the CCI.

$$\mathcal{F}_{\text{struct}}[\Phi] = \lambda_{\text{stab}}\, E_{\text{stab}} + \lambda_{\text{cons}}\, E_{\text{cons}} + \lambda_{\text{corr}}\, E_{\text{corr}} + \lambda_{\text{dyn}}\, E_{\text{dyn}}$$

4. **Paper 05 — Physical Grounding of the CCI**

   Tests connections between coarse-grained histogram activity and structural information. Spearman $r_s = 0.760$ ($p < 0.0001$) in discrete ensemble.

5. **Paper 06 — Continuous Ensemble Study**

   Strong correlation between CCI and structural free energy in a 160-trajectory ensemble:

$$r_s(\mathcal{C}_{\text{CCI}},\, \mathcal{F}_{\text{struct}}) = 0.878, \quad p < 10^{-50}$$

6. **Paper 07 — Entropy–Information Scaling (1D)**

   Empirical power-law scaling in 1D with phase diagrams:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}, \quad \alpha \approx 2.5, \quad r_s = 0.734, \quad p \approx 1.6 \times 10^{-21}$$

7. **Paper 08 — Landau-Type Theory**
   Proposes a Landau-type theoretical framework for structural selection. Order parameter $\mathcal{O} \sim I_{\text{nn}}$, control parameter $\mu \sim \mathcal{C}_{\text{CCI}}$:

$$\mathcal{F}_{\text{struct}}[\mathcal{O}] = a(\mu)\, \mathcal{O}^2 + b\, \mathcal{O}^4 + \gamma(\nabla \mathcal{O})^2, \quad a(\mu) = a_0(\mu - \mu_c)$$

   Theoretical proposal; Landau coefficients not yet derived from first principles.

8. **Paper 09 — Entropy–Information Scaling (2D)**
   Tests robustness of the scaling in two-dimensional ϕ^4 dynamics:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}, \quad \alpha \approx 3.0 \text{ (within tested range)}, \quad r_s \approx 0.87, \quad R^2 \approx 0.74$$

   Structural information I_{nn} is the dominant factor in 2D; histogram activity is subleading.

9. **Paper 10 — Dimensional Dependence of Scaling**
Comparative analysis of the scaling exponent across 1D and 2D, finding evidence for a dimension-dependent universality class:

$$\alpha_{1D} \approx 2.5 \longrightarrow \alpha_{2D} \gtrsim 3.0, \qquad r_s : 0.734 \to 0.870$$

The dimensional dependence of α is a property of the projection from $\mathcal{F}_{\text{struct}}$ onto the entropy–information plane.

10. **Paper 11 — Entropy–Information Scaling (3D)**
Extension to three-dimensional ϕ^4 dynamics reveals a breakdown of simple power-law scaling. Instead of a sharp exponent, a broad plateau emerges over $\alpha \in [2.8, 3.5]$, with significantly reduced regression power ($R^2 \approx 0.37$) but persistent monotonic correlation:

$$r_s \approx 0.85, \quad p \approx 1.6 \times 10^{-7}, \quad \alpha : \text{plateau } [2.8, 3.5]$$

The notion of a single effective exponent breaks down in 3D — a degeneracy of effective scaling descriptions emerges.

11. **Paper 12 — Cross-Dimensional Collapse Analysis**
Direct visual comparison of the scaling relation across 1D, 2D, and 3D via collapse plots, testing whether a unified cross-dimensional description exists. A partial collapse is observed: the monotonic structure is shared across dimensions, but no single curve provides a universal description.

Scaling survives, but the exponent does not.

12. **Paper 13 — Multi-Parameter Scaling Analysis**
Extends the single-exponent ansatz to a two-parameter model $\mathcal{C}_{\text{CCI}} \sim \dot{S}_{\text{cg}}^{(+)a} I_{\text{nn}}^{-b}$, finding systematic dimensional restructuring of the scaling weights:

$$\text{1D: } a \approx -1.6,\ b \approx 0.33,\ R^2 \approx 0.76 \qquad \text{2D: } a \approx -0.08,\ b \approx 0.50,\ R^2 \approx 0.74$$

$$\text{3D: } a \approx +0.30,\ b \approx 0.30,\ R^2 \approx 0.37$$

The entropy weight a changes sign from 1D to 3D, revealing a qualitative shift in the role of dynamical activity. *The exponent description fails not because the scaling breaks, but because the system transitions to a higher-dimensional multi-parameter scaling structure.*

Hierarchical Interpretation

A coherent three-level picture emerges across the series:

Level	Quantity	Role
Observable index	$\mathcal{C}_{\text{CCI}}$	Structural instability measure
Empirical proxy	$\dot{S}_{\text{cg}}^{(+)}/I_{\text{nn}}^{\alpha}$	Scaling component of CCI
Selection functional	$\mathcal{F}_{\text{struct}}$	Global effective description

And the three dynamical regimes:

Regime	I_{nn}	$\dot{S}_{\text{cg}}$
Ordered	high	low
Critical	moderate	moderate
Chaotic	low	high

The CCI plays a role analogous to Reynolds-type numbers in fluid and plasma systems: low CCI preserves coherent structures; high CCI drives the system toward chaotic, instability-dominated behaviour.

Future Work

- Extension to 4D and beyond; universality class determination
- Theoretical derivation of the multi-parameter scaling family $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)a}\, I_{\mathrm{nn}}^{-b}$ from the structural free energy
- Application to other field models (sine-Gordon, Kuramoto–Sivashinsky, reaction-diffusion)
- First-principles derivation of the scaling exponent α and its dimensional dependence
- Derivation of Landau coefficients from the microscopic field theory
- Connection to Lyapunov spectra, turbulence, and critical phenomena

Code & Data

All companion code is openly available:

https://github.com/Chris4081/structural-selection-principle